



PRECURSOR ANALYSIS REPORT: DOPPELPAYMER RANSOMWARE ATTACK ON PETROLEOS MEXICANOS (PEMEX) 2019

Cybersecurity for the Operational Technology
Environment (CyOTE)

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PRECURSOR ANALYSIS REPORT: DOPPELPAYMER RANSOMWARE ATTACK ON PETROLEOS MEXICANOS (PEMEX) 2019

1. EXECUTIVE SUMMARY

The DoppelPaymer Ransomware Attack on Petroleos Mexicanos (PEMEX) 2019 Precursor Analysis Report leverages publicly available information about the PEMEX cyber attack and catalogs anomalous observables for each technique employed in the attack. This analysis is based upon the methodology of the Cybersecurity for the Operational Technology Environment (CyOTE) program.

The 2019 DoppelPaymer ransomware attack on PEMEX, Mexico's nationalized petroleum corporation, highlights a unique threat that ransomware and cybercriminal extortion poses to Operational Technology (OT) environments in critical infrastructure. The incident began with an employee downloading commodity malware that allowed adversaries to gain initial access to PEMEX's enterprise environment. After conducting privilege escalation, tool ingress, and data exfiltration, the adversaries deployed DoppelPaymer ransomware throughout the PEMEX enterprise environment, resulting in the company having to take dozens of systems offline for at least several days. Although PEMEX stated that their operations were not affected, the data exfiltrated from PEMEX was made available for download on DoppelPaymer's leak site, as well as on other illicit criminal forums. This stolen data included not only company information, but also sensitive OT-specific configuration data. This incident showcases how cybercriminal exfiltration and posting of sensitive OT architecture documentation can pose security concerns for the targeted organization for years due to the long lifespan of OT assets and architectures.

Researchers and analysts identified 18 unique techniques utilized during the attack with a total of 190 observables using MITRE ATT&CK® for Industrial Control Systems. The CyOTE program assesses observables accompanying techniques used prior to the triggering event to identify opportunities to detect malicious activity. If observables accompanying the attack techniques are perceived and investigated prior to the triggering event, earlier comprehension of malicious activity can take place. Fifteen of the identified techniques used during the DoppelPaymer ransomware attack were precursors to the triggering event. Analysis identified 163 observables associated with these precursor techniques, 34 of which were assessed to have an increased likelihood of being perceived in the 60 days preceding the triggering event. The response and comprehension time could have been reduced if the observables had been identified earlier.

The information gathered in this report contributes to a library of observables tied to a repository of artifacts, data sources, and technique detection references for practitioners and developers to support the comprehension of indicators of attack. Asset owners and operators can use these products if they experience similar observables or to prepare for comparable scenarios.

2. INTRODUCTION

The [Cybersecurity for the Operational Technology Environment \(CyOTE\)](#) program developed capabilities for energy sector organizations to independently identify adversarial tactics and techniques within their operational technology (OT) environments. Led by Idaho National Laboratory (INL) under leadership of the Department of Energy (DOE) Office of Cybersecurity, Energy Security, and Emergency Response (CESER), CyOTE is a partnership with energy sector owners and operators whose goal is to tie the impacts of a cyber attack to anomalies in the OT environment to determine whether the anomalies have a malicious cyber cause.

2.1. APPLYING THE CYOTE METHODOLOGY

The CyOTE methodology, as shown in [Figure 1. CyOTE Methodology](#), applies fundamental concepts of perception and comprehension to a universe of knowns and unknowns increasingly disaggregated into observables, anomalies, and triggering events. The program utilizes MITRE's ATT&CK® Framework for Industrial Control Systems (ICS) as a common lexicon to assess triggering events. By leveraging the CyOTE methodology with existing commercial monitoring capabilities and manual data collection, energy sector partners can understand relationships between multiple observables, which could represent a faint signal of an attack requiring investigation. CyOTE can assist organizations in prioritizing their OT environment visibility investments.

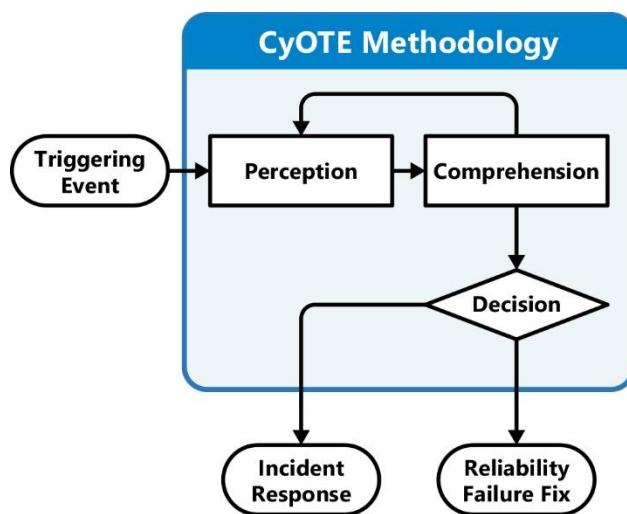


Figure 1. CyOTE Methodology

Historical case studies such as this one support continued learning through analysis of incidents that have impacted OT. This precursor analysis report is based on publicly available reports and provides examples of how key concepts in the CyOTE methodology appear in the real world, providing insights on how similar novel attacks could be detected earlier and therefore mitigated. The analysis enables OT personnel to independently identify observables associated with techniques known to be indicators of attack within OT environments. The identified observables highlight anomalous events for further investigation, which could enhance comprehension of malicious activity.

A timeline of events based on the CyOTE methodology portrays the attack-related observables associated with the precursor analysis report cyber attack. The timeline includes assessed dates, the triggering event, and comprehension of malicious activity by the organization. The point on this timeline when each technique appears is critical to the organization's ability to perceive and comprehend the associated malicious activity. Perception of techniques early in the timeline is critical, since halting those techniques will generally have greater potential to limit additional attack vectors using other techniques, defeat the cyber attack, and limit damage to operations.

Each technique has an assessed perceivability. Perceivability is a function of the number of observables and the potential for personnel to detect those observables. If a technique includes effects which personnel may detect, such as deletion or modification of system files or required user execution, then the technique would be more perceivable.

Differences in infrastructure and system configurations may present different challenges and opportunities for observable detection. For example, architecture-wide endpoint monitoring is likely to improve the perceivability of techniques which modify host files, such as the Data Destruction technique (T0809) for Inhibit Response Function and Theft of Operational Information technique (T0882) for Impact. Network monitoring and log analysis capabilities are likely to improve perceivability of techniques which create malicious network traffic, such as the Standard Application Layer Protocol technique (T0869) for Command and Control, External Remote Services technique (T0822) for Initial Access, and Connection Proxy technique (T0884) for Command and Control. Alternatively, enhancing the monitoring parameters of system files would increase the perceivability of techniques such as Data from Information Repositories technique (T0811) for Collection and the Service Stop technique (T0881) for Inhibit Response Function.

Comprehension can be further enhanced by technique artifacts created when adversaries employ certain attack techniques. The CyOTE program provides organizations with a [library of observables](#) reported in each historical case. The library can be used in conjunction with a repository of artifacts, data sources, and technique detection references for practitioners and developers to support the comprehension of indicators of attack.

2.2. BACKGROUND ON THE ATTACK

The 2019 DoppelPaymer ransomware attack on Petroleos Mexicanos (PEMEX), Mexico's nationalized petroleum company, began with an employee downloading commodity malware that is often used for initial access, such as Emotet or Dridex, likely sometime in early September 2019 (D-60).^{1,a} During the same period adversaries may have also used the FakeUpdates malware framework, a drive-by compromise masquerading as a browser update, to gain initial access.

After achieving initial access and profiling the victim environment, the adversaries proceeded to escalate privileges, exfiltrate data, and move additional tools and payloads into the PEMEX network. Adversaries eventually deployed the DoppelPaymer ransomware on 10 November (D-0).² Numerous PEMEX enterprise systems displayed a ransom note demanding payment of 565 Bitcoin (then equivalent to \$4,899,295.80 USD).³ On 11 November (D+1), PEMEX issued a statement that operation and production systems were not compromised and that the attack was neutralized.⁴ Although PEMEX stated that OT systems were unaffected by the ransomware incident, public reporting indicated that PEMEX employees were not able to access their enterprise systems for days after the attack. Further, the data exfiltrated from PEMEX was available for download on DoppelPaymer's leak site, as well as on other illicit criminal forums. This stolen data included not only company information, but also sensitive OT-specific configuration data.

A timeline of adversarial techniques is shown in [Error! Reference source not found.](#) The timeline includes the estimated number of days prior to and after the triggering event. The timeline after the triggering event includes the assessed victim comprehension timeline.

After gaining initial access sometime in early September 2019, the adversaries proceeded to escalate privileges, move laterally, and exfiltrate data from PEMEX from D-60 until around D-1. The exfiltration of sensitive OT environment documentation is a significant concern: according to cybersecurity firm Mandiant, one in every seven critical infrastructure ransomware incidents involves the leak of sensitive OT data.⁵ Such data stored in the enterprise environment may be subject to inadvertent exfiltration by a ransomware group(s) as part of their strategy to pressure the victim into paying a ransom. The attack on PEMEX involved several groups of cybercriminal adversaries, including initial access brokers, botnet masters, and ransomware operators.

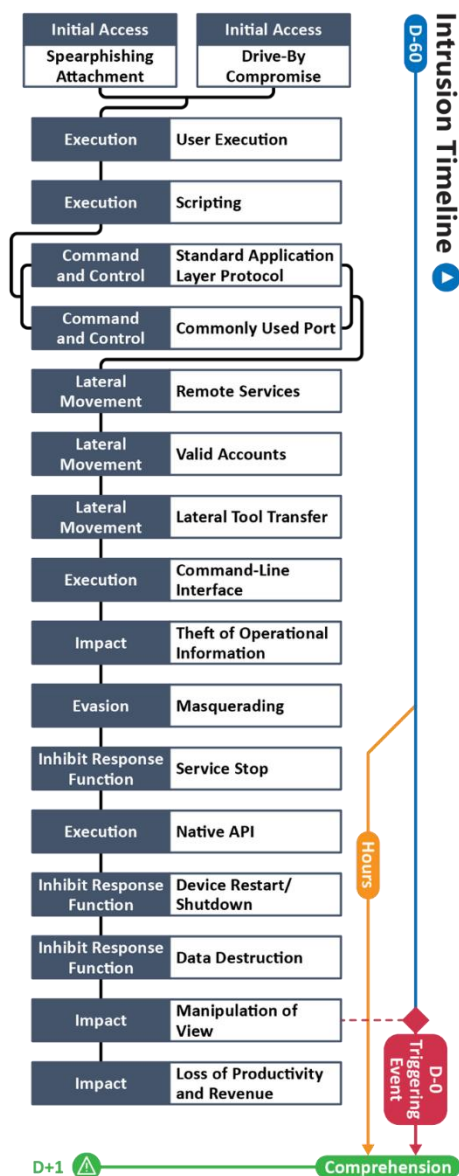


Figure 2. Intrusion Timeline

^a IBM stated the average time from initial access to ransom demand for DoppelPaymer attacks in 2019 was over two months.

Analysis identified 18 unique techniques in a sequence and timeframe likely used by adversaries during this cyber attack ([Error! Reference source not found. 1](#)). These attack techniques are defined according to MITRE's ATT&CK® for ICS framework.

Table 1. Techniques Used in the DoppelPaymer Ransomware Attack on PEMEX in 2019

Initial Access	Execution	Persistence	Privilege Escalation	Evasion	Discovery	Lateral Movement	Collection	Command and Control	Inhibit Response Function	Impair Process Control	Impact
Data Historian Compromise	Change Operating Mode	Modify Program	Exploitation for Privilege Escalation	Change Operating Mode	Network Connection Enumeration	Default Credentials	Automated Collection	Commonly Used Port	Activate Firmware Update Mode	Brute Force I/O	Damage to Property
Drive-by Compromise	Command-Line Interface	Module Firmware	Hooking	Exploitation for Evasion	Network Sniffing	Exploitation of Remote Services	Data from Information Repositories	Connection Proxy	Alarm Suppression	Modify Parameter	Denial of Control
Engineering Workstation Compromise	Execution through API	Project File Infection		Indicator Removal on Host	Remote System Discovery	Lateral Tool Transfer	Detect Operating Mode	Standard Application Layer Protocol	Block Command Message	Module Firmware	Denial of View
Exploit Public-Facing Application	Graphical User Interface	System Firmware		Masquerading	Remote System Information Discovery	Program Download	I/O Image		Block Reporting Message	Spoof Reporting Message	Loss of Availability
Exploitation of Remote Services	Hooking	Valid Accounts		Rootkit	Wireless Sniffing	Remote Services	Man in the Middle		Block Serial COM	Unauthorized Command Message	Loss of Control
External Remote Services	Modify Controller Tasking			Spoof Reporting Message		Valid Accounts	Monitor Process State		Data Destruction		Loss of Productivity and Revenue
Internet Accessible Device	Native API						Point & Tag Identification		Denial of Service		Loss of Protection
Remote Services	Scripting						Program Upload		Device Restart/Shutdown		Loss of Safety
Replication Through Removable Media	User Execution						Screen Capture		Manipulate I/O Image		Loss of View
Rogue Master							Wireless Sniffing		Modify Alarm Settings		Manipulation of Control
Spearphishing Attachment									Rootkit		Manipulation of View
Supply Chain Compromise									Service Stop		Theft of Operational Information
Transient Cyber Asset									System Firmware		
Wireless Compromise											

Table 2. Precursor Analysis Report Quantitative Summary

Precursor Analysis Report Quantitative Summary	Totals
MITRE ATT&CK® for ICS Techniques	18
Technique Observables	190
Precursor Techniques	15
Precursor Technique Observables	163
Highly Perceivable Precursor Technique Observable	34

3. OBSERVABLE AND TECHNIQUE ANALYSIS

The following analysis may assist organizations in identifying malicious cyber activity earlier and more effectively. The following techniques and observables were compiled from publicly available sources and correlated with expert analysis.

3.1. SPEARPHISHING ATTACHMENT TECHNIQUE (T0865) FOR INITIAL ACCESS

Adversaries likely used malicious spearphishing attachment campaigns to facilitate the download and execution of Emotet, Dridex, or both. Emotet and Dridex are banking trojans that often act as initial access vectors for ransomware groups. Ransomware groups buy access from botnet operators and then conduct further malicious activity in a victim's environment, including ransomware deployment.⁶ Adversaries who control banking trojans such as Emotet and Dridex regularly make use of Microsoft Office documents in malicious spam campaigns, with common email themes like finance or payroll to bait the end-user into interacting with the malicious attachment.⁷ In some cases, the Dridex malware is embedded in the document itself, not requiring an external resource to install the initial payload.⁸

IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management personnel may have been able to observe initial malicious email attachments before interacting with them. It is unclear in the case of PEMEX who may have initially interacted with the malicious attachment, but any individual with a company email is susceptible to this tactic.

A total of seven observables were identified with the use of the [Spearphishing Attachment technique \(T0865\)](#). This technique is important for investigation because it is a common method for adversaries to gain initial access into victim environments. This technique appears early in the timeline and responding to it will likely halt future events. Terminating the chain of techniques at this point would limit initial access vectors into the victim environment.

Of the seven observables associated with this technique, none are assessed to be highly perceivable. They are listed in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 29 artifacts could be generated by the Spearphishing Attachment technique
Technique Observers^b	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, Management
Resources	Technique Detection References

^b Observer titles are adapted from the Job Role Groupings listed in [the SANS ICS Job Role to Competency Level Poster](#). CyOTE products utilize these job categories rather than organizational titles to both support comprehensive analysis and preserve anonymity within the victim organization. A complete list of potential observers can be found in [Appendix C](#).

3.2. DRIVE-BY COMPROMISE TECHNIQUE (T0817) FOR INITIAL ACCESS

Adversaries may have also leveraged the Drive-by Compromise technique (T0817) to gain initial access into the PEMEX enterprise network. Dridex operators may have made use of a malware framework known as FakeUpdates or SocGhoulish during of the compromise. FakeUpdates was observed downloading Dridex in October and November of 2019, several weeks before the attack on PEMEX.⁹ FakeUpdates, as the name implies, is malware that masquerades as Google Chrome, FireFox, Opera, or Internet Explorer browser updates that redirect a user to a malicious site and prompts them to download a “browser update.” Because FakeUpdates makes use of Search Engine Optimization (SEO) poisoning, it is likely a PEMEX employee used a browser to search for a targeted subject and then interacted with a malicious URL indexed by that search result. Once installed, FakeUpdates can then download other malware payloads such as Dridex. After the compromised site redirects the victim user, the user is prompted to download the malware, often written in Hypertext Markup Language (HTML), JavaScript (.js), or a ZIP file containing a JavaScript file.¹⁰

IT Staff, IT Cybersecurity, Engineering, Support Staff, and Management personnel may have been able to observe compromised websites prompting them to download a file masquerading as a browser update, as well as the downloading of files onto the host.

A total of seven observables were identified with the use of the [Drive-by Compromise technique \(T0817\)](#). This technique is important for investigation because it results in the download of malicious software into a victim’s enterprise environment. This technique appears early in the timeline and responding to it will likely prevent initial access. Terminating the chain of techniques at this point may limit further adversary activity in the victim environment and prevent the theft of operational information.

Of the seven observables associated with this technique, none are assessed to be highly perceivable. They are listed in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 24 artifacts could be generated by the Drive-by Compromise technique
Technique Observers	IT Staff, IT Cybersecurity, Engineering, Support Staff, Management
Resources	Technique Detection References

3.3. USER EXECUTION TECHNIQUE (T0836) FOR EXECUTION

Adversaries leveraged one or more unwitting victim employees to deploy first stage malware such as Emotet, Dridex, or FakeUpdates via the User Execution technique (T0836). Victims are prompted to enable malicious Macros in Microsoft Office documents that load either Dridex or Emotet malware onto the victim's host machine. These documents prompt the victim to enable macros, which then execute embedded code in the document that reaches out to an external resource to download a malware payload. External resources like compromised third-party websites often host an initial payload of Emotet, Dridex, or other malware. Dridex is also capable of being loaded directly into memory from malicious Office documents, negating the need for outbound network communications for an initial download.¹¹ Victim users may also have downloaded the FakeUpdates malware framework, believing it to be a legitimate browser update.

IT Staff, IT Cybersecurity, Engineering, Support Staff, and Management personnel may have been able to observe the malicious Office Documents or the faux browser update prompts that downloaded FakeUpdates.

A total of 13 observables were identified with the use of the [User Execution technique \(T0836\)](#). This technique is important for investigation because it is one of the most common gateways cyber adversaries utilize to gain initial access. This technique appears early in the timeline and responding to it will likely halt all future adversary activity in the victim's environment. Terminating the chain of techniques at this point would effectively halt all further adversary activity and limit impact to business operations.

Of the 13 observables associated with this technique, none are assessed to be highly perceivable. They are listed in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 23 artifacts could be generated by the User Execution technique
Technique Observers	IT Staff, IT Cybersecurity, Engineering, Support Staff, Management
Resources	Technique Detection References

3.4. SCRIPTING TECHNIQUE (T0853) FOR EXECUTION

The adversaries made use of the Scripting technique (T0853) throughout the course of the PEMEX intrusion. Malicious documents that download Emotet or Dridex often abuse Visual Basic for Applications (VBA) when an end-user enables Macros to either download a malware payload or load malicious code directly into memory. Additionally, the FakeUpdates malware has been observed using both JavaScript and Hypertext Markup Language (HTML) files to install the initial module that masquerades as a browser update inside of a .ZIP file.¹²

IT Staff and IT Cybersecurity personnel may have been able to observe malicious documents and prompts for a fake browser update associated with malware utilizing this technique.

A total of 20 observables were identified with the use of the [Scripting technique \(T0853\)](#). This technique is important for investigation because it allows the adversary to conduct malicious actions in a victim's environment, often facilitating initial access or lateral movement. This technique appears relatively early in the timeline and responding to it will likely halt further adversary activity within the victim's environment. Terminating the chain of techniques at this point would limit malicious activity in the victim's environment, as well as avert future events such as theft of operational information and manipulation of view.

Of the 20 observables associated with this technique, none are assessed to be highly perceivable. They are listed in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 12 artifacts could be generated by the Scripting technique
Technique Observers	IT Staff, IT Cybersecurity
Resources	Technique Detection References

3.5. STANDARD APPLICATION LAYER PROTOCOL TECHNIQUE (T0869) FOR COMMAND AND CONTROL

The adversaries used several standard application layer protocols throughout the intrusion for command and control (C2), tool ingress, and data exfiltration. Malware like Emotet, Dridex, and FakeUpdates use HTTP and HTTPS for routine C2 communications, as well as for fetching other payloads from external resources. Additionally, adversaries almost certainly used File Transfer Protocol (FTP) and Secure File Transfer Protocol (SFTP) to move additional tools and malware payloads into PEMEX's environment, as well as to exfiltrate company data later listed on the DoppelPaymer leak site.

IT Staff and IT Cybersecurity personnel may have been able to observe C2 traffic associated with Emotet, Dridex, and FakeUpdates. Furthermore, IT Staff and IT cybersecurity may have been able to observe the data exfiltration traffic, which in the case of the PEMEX attack may have lasted several weeks due to the volume of data involved.

A total of 10 observables were identified with the use of the [Standard Application Layer Protocol technique \(T0869\)](#). This technique is important for investigation as prolonged anomalous network traffic is a strong indication of adversary activity. This technique appears throughout the timeline and responding to it will alert defenders to malicious activity within their environment. Terminating the chain of techniques at this point would likely halt any further adversary activity if defenders took steps to block the malicious network traffic.

Of the 10 observables associated with this technique, four are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 12 artifacts could be generated by the Standard Application Layer Protocol technique
Technique Observers	IT Staff, IT Cybersecurity
Resources	Technique Detection References

3.6. COMMONLY USED PORT TECHNIQUE (T0855) FOR COMMAND AND CONTROL

The adversaries employed commonly used ports throughout the intrusion for C2, tool ingress, and data exfiltration. Emotet and Dridex commonly use TCP Ports 80, 443, and 8080 for C2, although both also have been observed using non-standard ports. FakeUpdates also regularly makes use of Ports 80 and 443 for download and C2. Adversaries likely used Ports 21 and 22 for data exfiltration while using Rclone or Mega, which are utilities designed for data transfer and cloud storage, respectively.

IT Staff and IT Cybersecurity personnel may have been able to observe the anomalous traffic to and from C2 servers, as well as outbound data exfiltration traffic at odd hours.

A total of 10 observables were identified with the use of the [Commonly Used Port technique \(T0855\)](#). This technique is important for investigation as port usage is an indicator of adversary activity in the victim's environment. This technique appears throughout the timeline and responding to it may halt future activity. Terminating the chain of techniques at this point would limit adversary activity in the victim's environment and prevent communication with malware already present.

Of the 10 observables associated with this technique, four are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 5 artifacts could be generated by the Commonly Used Port technique
Technique Observers	IT Staff, IT Cybersecurity
Resources	Technique Detection References

3.7. REMOTE SERVICES TECHNIQUE (T0886) FOR LATERAL MOVEMENT

The adversaries likely employed the Remote Services technique (T0886) to facilitate lateral movement within the PEMEX enterprise environment. Financially motivated cyber threat actors often abuse native remote services, such as Server Message Block (SMB) or Remote Desktop Protocol (RDP), with valid accounts to conduct further reconnaissance and lateral movement. This often leads to critical assets such as domain controllers being compromised and then used as a springboard for enterprise-wide deployment of ransomware.

IT Staff and IT Cybersecurity personnel may have been able to observe the anomalous network traffic related to the SMB or RDP protocols at irregular hours.

A total of four observables were identified with the use of the [Remote Services technique \(T0886\)](#). This technique is important for investigation as it is often leveraged to facilitate lateral movement in a victim's environment, allowing for further malicious activity. This technique appears throughout the timeline and responding to it may halt future activity. Terminating the chain of techniques at this point would limit adversary activity in the victim's environment and prevent communication with installed malware that acts as a foothold in a victim's environment.

Of the four observables associated with this technique, two are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 24 artifacts could be generated by the Remote Services technique
Technique Observers	IT Staff, IT Cybersecurity
Resources	Technique Detection References

3.8. VALID ACCOUNTS TECHNIQUE (T0859) FOR LATERAL MOVEMENT

Financially motivated adversaries often make use of stolen credentials for valid accounts that precursor malware, such as Dridex, can harvest from web browsers and email clients on infected hosts. The adversaries may have used stolen credentials from an initial commodity malware infection for further discovery and lateral movement in the victim's environment.

IT Staff and IT Cybersecurity personnel may have been able to observe logons from valid user credentials at anomalous hours.

A total of six observables were identified with the use of the [Valid Accounts technique \(T0859\)](#). This technique is important for investigation because adversaries often use stolen credentials to help facilitate lateral movement in a victim's environment. This technique appears early in the timeline and responding to it will limit an adversary's ability to persist or move laterally in a victim's environment via bypassing access controls. Terminating the chain of techniques at this point would partially limit an adversary's presence in the environment, as these credentials may have been harvested by malware still present in the system.

Of the six observables associated with this technique, two are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 16 artifacts could be generated by the Valid Accounts technique
Technique Observers	IT Staff, IT Cybersecurity
Resources	Technique Detection References

3.9. LATERAL TOOL TRANSFER TECHNIQUE (T0867) FOR LATERAL MOVEMENT

Adversaries utilized the Lateral Tool Transfer technique (T0867) throughout the intrusion. Upon gaining initial access, adversaries may have used banking malware such as Emotet to download and execute Dridex within the PEMEX enterprise environment, allowing a separate group of adversaries access. From there, the adversaries proceeded to deploy additional tools, such as Mimikatz, Koadic, and PoshC2 to help elevate privileges and move laterally, between 30 minutes and two hours after initial Dridex execution.¹³ The adversaries likely also transferred data exfiltration tools, such as Rclone or Mega, into the victim's environment. These tools are used to steal sensitive company data from victims over a period of several weeks to facilitate double extortion, a popular tactic ransomware adversaries use to strong-arm victims into paying ransoms to avoid legal penalties due to violation of data privacy laws.

IT Staff and IT cybersecurity personnel may have been able to observe the presence of anomalous files as well as the anomalous network traffic associated with the download of additional tools and payloads.

A total of 22 observables were identified with the use of the [Lateral Tool Transfer technique \(T0867\)](#). This technique is important for investigation because it indicates adversaries are moving additional tools and payloads into the victim environment to perform further malicious activity. This technique appears throughout the timeline and responding to it will limit further adversary activity. Terminating the chain of techniques at this point would prevent ransomware deployment and data exfiltration from the enterprise environment.

Of the 22 observables associated with this technique, eight are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 22 artifacts could be generated by the Lateral Tool Transfer technique
Technique Observers	IT Staff, IT Cybersecurity
Resources	Technique Detection References

3.10. COMMAND-LINE INTERFACE TECHNIQUE (T0807) FOR EXECUTION

Adversaries utilized the Command-Line Interface technique (T0807) throughout the timeline. This technique most likely would have been associated with the adversaries executing PowerShell commands, Active Directory (AD) reconnaissance commands, or using Command-Line Interface (CLI)-based tools to exfiltrate data. DoppelPaymer is designed to execute only after a specific command-line argument is provided to hamper analysis and reverse engineering of the malware.

IT Staff and IT Cybersecurity personnel may have been able to observe various command-line executions associated with PowerShell, AD reconnaissance, or data exfiltration tools.

A total of 16 observables were identified with the use of the [Command-Line Interface technique \(T0807\)](#). This technique is important for investigation because it often is associated with adversaries executing malicious payloads within the victim's environment or conducting reconnaissance. This technique appears throughout the timeline and responding to it may prevent the adversaries from deploying or executing the ransomware. Terminating the chain of techniques at this point would prevent prolonged data exfiltration.

Of the 16 observables associated with this technique, four are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 25 artifacts could be generated by the Command-Line Interface technique
Technique Observers	IT Staff, IT Cybersecurity
Resources	Technique Detection References

3.11. THEFT OF OPERATIONAL INFORMATION TECHNIQUE (T0882) FOR IMPACT

A common tactic that ransomware operators use is known as “double extortion,” which entails exfiltrating victim data prior to deploying the ransomware payload, and then threatening to post the stolen data to pressure the victim into paying a ransom.¹⁴ The adversaries often use tools such as Mega and Rclone, which employ FTP and SFTP, to accomplish large-scale data exfiltration up to several weeks prior to ransomware deployment. The time required to exfiltrate large volumes of data provides a considerable window of opportunity for energy sector and other critical infrastructure organizations to identify this activity.

IT Staff and IT Cybersecurity personnel may have been able to observe prolonged network traffic to an external server associated with data exfiltration.

A total of 17 observables were identified with the use of the [Theft of Operational Information technique \(T0882\)](#). This technique is important for investigation because theft of operational data jeopardizes not only business practices but the security of the OT environment. This technique appears late in the timeline and responding to it will prevent the adversaries from exfiltrating data from the enterprise and OT environments. Terminating the chain of techniques at this point would limit operational damage and the loss of sensitive OT documentation.

Of the 17 observables associated with this technique, five are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 4 artifacts could be generated by the Theft of Operational Information technique
Technique Observers	IT Staff, IT Cybersecurity
Resources	Technique Detection References

3.12. MASQUERADING TECHNIQUE (T0849) FOR EVASION

DoppelPaymer ransomware makes use of stolen signed software certificates to evade signature-based detection engines and ensure stealthy execution. DoppelPaymer uses stolen software certificates and attempts to masquerade as the “SpotLife WebAlbum Service Plugin” developed by Logitech.¹⁵

IT Cybersecurity personnel are unlikely to have observed the stolen software certificates masquerading as Logitech plugins due to the extremely common use of Logitech-associated certificates.

One observable was identified with the use of the [Masquerading technique \(T0849\)](#). This technique is important for investigation because it circumvents critical security tools that can alert a victim of malicious cyber activity. This technique appears late in the timeline and responding to it will likely halt the execution of the malware, although this is unlikely given the brief time from disabling services to ransomware execution. Terminating the chain of techniques at this point would halt the deployment of the ransomware.

The one observable associated with this technique is not assessed to be highly perceivable. It is listed in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 15 artifacts could be generated by the Masquerading technique
Technique Observers	IT Cybersecurity
Resources	Technique Detection References

3.13. SERVICE STOP TECHNIQUE (T0881) FOR INHIBIT RESPONSE FUNCTION

DoppelPaymer attempts to disable various processes once executed in the victim's environment. The ransomware makes use of Process Hacker to identify and terminate any listed processes hardcoded in its configuration.¹⁶ Any blacklisted processes are identified via DoppelPaymer using Process Hacker to check the CRC32 hash of the specific process. If a process hash matches to a blacklisted hash, DoppelPaymer then leverages Process Hacker to open a handle to that process and kill it.¹⁷ These processes include common Windows security applications, as well as commercial anti-virus products.

IT Cybersecurity personnel may have been able to observe the malfunctioning or nonfunctioning of Windows security programs or the execution of DoppelPaymer. However, this would be after the deployment and execution of DoppelPaymer throughout the enterprise environment, so any observer attempts to halt execution at this point are unlikely to succeed.

A total of 14 observables were identified with the use of the [Service Stop technique \(T0881\)](#). This technique is important for investigation because it disables critical security tools that can alert a victim to malicious cyber activity. This technique appears late in the timeline and responding to it would likely halt final execution of the ransomware, although it is highly unlikely defenders would have sufficient time to act. Terminating the chain of techniques at this point would limit operational damage.

Of the 14 observables associated with this technique, four are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 13 artifacts could be generated by the Service Stop technique
Technique Observers	IT Cybersecurity
Resources	Technique Detection References

3.14. NATIVE API TECHNIQUE (T0834) FOR EXECUTION

Commodity malware often makes use of the Native API technique (T0834) to help evade behavior-based antivirus engines and interactions with the Windows OS. DoppelPaymer uses the Native API technique (T0834) while halting and disabling security tools before execution via the ZwTerminateProcess Application Programming Interface (API) call.¹⁸ DoppelPaymer also uses the NtCreateFile and ZwDeleteFile API calls.

IT Cybersecurity personnel may have been able to observe execution of Native Windows OS APIs via behavior-based detection engines.

A total of 14 observables were identified with the use of the [Native API technique \(T0834\)](#). This technique is important for investigation because adversaries often use native APIs in malware to avoid behavior-based antivirus engines. This technique appears late in the timeline and responding to it may halt execution of the malware, although it is highly unlikely defenders would have sufficient time to act. Terminating the chain of techniques at this point would potentially halt the execution of DoppelPaymer.

Of the 14 observables associated with this technique, none are assessed to be highly perceivable. They are listed in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 25 artifacts could be generated by the Native API technique
Technique Observers	IT Cybersecurity
Resources	Technique Detection References

3.15. DEVICE RESTART/SHUTDOWN TECHNIQUE (T0816) FOR INHIBIT RESPONSE FUNCTION

DoppelPaymer changes user passwords before forcing a system restart into safe mode to prevent users from accessing the system. The ransomware will then change the notice text that appears before Windows proceeds to the login screen. Use of this technique may have “[affected] the operation of less than 5% of personal computer equipment,” as PEMEX reported in an official announcement.¹⁹

IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management personnel may have been able to observe anomalous host shutdowns and restarts.

A total of two observables were identified with the use of the [Device Restart/Shutdown technique \(T0816\)](#). This technique is important for investigation because anomalous shutdowns often are part of the final stage of an enterprise-wide ransomware attack. Anomalous shutdowns may also indicate a reliability incident with malfunctioning equipment. This technique appears late in the timeline and responding to it may halt the execution and spread of DoppelPaymer. Terminating the chain of techniques at this point could halt the execution of the ransomware, although it is highly unlikely defenders would have sufficient time to act.

Of the two observables associated with this technique, one is assessed to be highly perceivable. It is italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 17 artifacts could be generated by the Device Restart/Shutdown technique
Technique Observers	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, Management
Resources	Technique Detection References

3.16. DATA DESTRUCTION TECHNIQUE (T0809) FOR INHIBIT RESPONSE FUNCTION

Upon execution of DoppelPaymer, the malware encrypts and appends any targeted file extensions with the .doppel file extension. This effectively renders any targeted files unusable by end-users. Infected hosts would also display a ransom note on enterprise systems informing users that their files were encrypted, and the only way to decrypt them was to pay the ransom.

IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management likely observed files with the .doppel file extension and were unable to access common file types.

A total of 17 observables were identified with the use of the [Data Destruction technique \(T0809\)](#). This technique is important for investigation because it renders files crucial to business and other enterprise operations unusable. This technique appears late in the timeline and responding to it at this point in the timeline is unlikely to minimize the impact of DoppelPaymer ransomware deployment.

Of the 17 observables associated with this technique, five are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 27 artifacts could be generated by the Data Destruction technique
Technique Observers	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, Management
Resources	Technique Detection References

3.17. MANIPULATION OF VIEW TECHNIQUE (T0832) FOR IMPACT

Upon execution of DoppelPaymer, the malware would encrypt any targeted file extensions with the .dopeled extension and display a ransom note on enterprise systems informing users that their files were encrypted and the only way to decrypt them was to pay the ransom.

IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management likely observed the ransom note on infected hosts in the enterprise environment.

A total of seven observables were identified with the use of the [Manipulation of View technique \(T0832\)](#). This technique is important for investigation because it prevents the organization from viewing the state of – and prevents users from interacting with – any compromised systems. This technique appears late in the timeline and represents the triggering event for the attack on PEMEX. Responding to it would include efforts to regain operational functionality and resume normal operation. Terminating the chain of techniques at this point would not limit destruction or business impacts.

Of the seven observables associated with this technique, five are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 5 artifacts could be generated by the Manipulation of View technique
Technique Observers	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, Management
Resources	Technique Detection References

3.18. LOSS OF PRODUCTIVITY AND REVENUE TECHNIQUE (T0828) FOR IMPACT

Although PEMEX released an official statement saying that fewer than 5% of their computers had been affected and the core company functions had not been impacted, PEMEX employees indicated that “entire floors of computers were wiped out” employees had to use old machines to “half-way work”.^{20,21} PEMEX also was forced to resort to manual billing operations, hindering business and payroll functions. Although the DoppelPaymer attack did not affect core PEMEX functions such as fuel production or supply operations, the impact of an enterprise-centric ransomware attack can impact normal business and supply chain operations.

IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management personnel likely observed the switch to manual business and payroll operations due to multiple hosts being rendered unusable by the DoppelPaymer ransomware.

A total of three observables were identified with the use of the [Loss of Productivity technique \(T0828\)](#). This technique is important for investigation to determine the extent of potential damage to systems and business losses. This technique appears after the triggering event and occurs beyond the point at which the victim could limit the impact of the attack.

Of the three observables associated with this technique, all are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 5 artifacts could be generated by the Loss of Productivity or Revenue technique
Technique Observers	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, Management
Resources	Technique Detection References

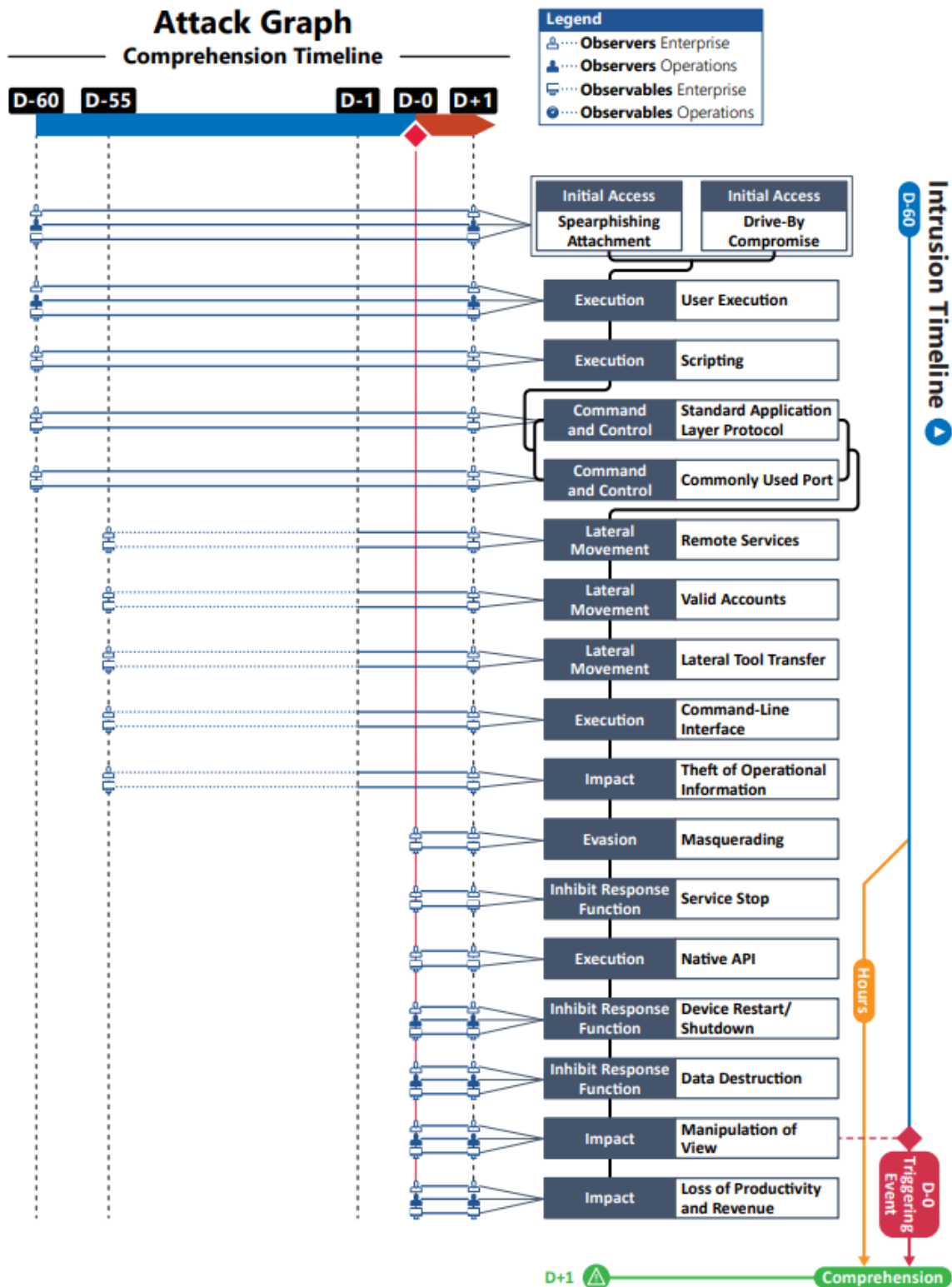


Figure 3. Attack Graph

APPENDIX A: OBSERVABLES LIBRARY

NOTE: Highly perceivable observables are italicized and marked †.

Observables Associated with Spearphishing Attachment Technique (T0865)	
Observable 1	Presence of Anomalous Email Containing an Attachment: Office Document: Word Document
Observable 2	Presence of Anomalous Email Containing an Attachment: Office Document: Excel Document
Observable 3	Presence of Anomalous Email Containing an Attachment: Office Document: PDF Document
Observable 4	Presence of Anomalous Email Containing an Attachment: JAR File
Observable 5	Anomalous Network Traffic: Outbound: HTTP: Over TCP Port 80: HTTP GET Request
Observable 6	Anomalous Network Traffic: Outbound: HTTPS: Over TCP Port 443: HTTPS Get Request
Observable 7	Anomalous Prompt to Enable Macros

Observables Associated with Drive-By Compromise Technique (T0817)	
Observable 1	User Interaction with Anomalous Browser Update: Hypertext Markup Language File
Observable 2	User Interaction with Anomalous Browser Update: JavaScript File
Observable 3	User Interaction with Anomalous Browser Update: Compressed File: ZIP File
Observable 4	Anomalous Network Traffic: Inbound: HTTP: Over TCP Port 80: HTTP GET Request
Observable 5	Anomalous Network Traffic: Inbound: HTTPS: Over TCP Port 443: HTTP GET Request
Observable 6	Anomalous Network Traffic: Outbound: HTTPS: Over TCP Port 443: HTTP GET Request
Observable 7	Anomalous Network Traffic: Outbound: HTTPS: Over TCP Port 80: HTTP GET Request

Observables Associated with User Execution Technique (T0836)	
Observable 1	User Interaction with Anomalous Browser Update: Hypertext Markup Language File
Observable 2	User Interaction with Anomalous Browser Update: JavaScript File
Observable 3	User Interaction with Anomalous Browser Update: Compressed File: ZIP File
Observable 4	User Interaction with Anomalous Email: Opens Attachment: Containing Macros: Excel Document
Observable 5	User Interaction with Anomalous Email: Opens Attachment: Containing Macros: Word Document

Observables Associated with User Execution Technique (T0836)	
Observable 6	User Interaction with Anomalous Email: Opens Attachment: Containing Macros: PDF
Observable 7	User Interaction with Anomalous Email: Opens Attachment: JAR
Observable 8	Anomalous Network Traffic: Inbound: HTTP: Over TCP Port 80: HTTP GET Request
Observable 9	Anomalous Network Traffic: Inbound: HTTPS: Over TCP Port 443: HTTP GET Request
Observable 10	Anomalous Network Traffic: Outbound: HTTPS: Over TCP Port 443: HTTP GET Request
Observable 11	Anomalous Network Traffic: Outbound: HTTPS: Over TCP Port 80: HTTP GET Request
Observable 12	Presence of Anomalous Binary on Host: C:\Users\<username>\Desktop\p1q135no.exe
Observable 13	Presence of Anomalous Binary on Host: C:\Users\<username>\Desktop\DoppelPaymer.exe

Observables Associated with Scripting Technique (T0853)	
Observable 1	Anomalous Script Execution on Local Host: PowerShell
Observable 2	Anomalous Script Execution on Local Host: Visual Basic for Applications (VBA)
Observable 3	Anomalous Script Execution on Local Host: JavaScript
Observable 4	Anomalous Script Execution on Local Host: Hypertext Markup Language (HTML)
Observable 5	Anomalous Network Traffic: Inbound from External IP to Local Host: HTTP: Over TCP Port 80: HTTP GET Request
Observable 6	Anomalous Network Traffic: Inbound from External IP to Local Host: HTTPS: Over TCP Port 443: HTTP GET Request
Observable 7	Anomalous Network Traffic: Outbound from Local Host to External IP: Over Hypertext Transfer Protocol Secure TCP Port 443: HTTP GET Request
Observable 8	Anomalous Network Traffic: Outbound: HTTP: Over TCP Port 80: HTTP GET Request
Observable 9	UAC Window Pops Up
Observable 10	Presence of Anomalous Binary on Host
Observable 11	Presence of Anomalous Binary on Host: C:\Users\<username>\Desktop\p1q135no.exe
Observable 12	Presence of Anomalous Binary on Host: C:\Users\<username>\Desktop\DoppelPaymer.exe
Observable 13	Presence of Anomalous Binary on Host: Signed Executable: Spoofed signature: Logitech Plug-In: SpotLife WebAlbum Service Plugin
Observable 14	Presence of Anomalous File on Host: Anomalous Alternative Data Stream (ADS): In %AppData%

Observables Associated with Scripting Technique (T0853)

Observable 15	Anomalous Command-Line: 'C:\Users\<USER>\AppData\Roaming\<random>:<random> QWD5MRg95gUEfGVsvUGBY84h C:\Users\gratemin\Desktop\p1q135no.exe'
Observable 16	Anomalous Command Line: C:\Windows\system32\takeown.exe /F <service_name>
Observable 17	Anomalous Command Line: C:\Windows\system32\icacls.exe <service_name> /reset
Observable 18	Execution of Anomalous Binary on Host: C:\Users\<username>\Desktop\p1q135no.exe
Observable 19	Execution of Anomalous Binary on Host: Locator.exe
Observable 20	Creation of Anomalous Service on Host: RPC Locator: Locator.exe

Observables Associated with Standard Application Layer Protocol Technique (T0869)

Observable 1	Anomalous Network Traffic: From Local Host to External IP: Over Hypertext Transfer Protocol (HTTP) TCP Port 80: HTTP GET Request
Observable 2	Anomalous Network Traffic: From Local Host to External IP: Over Hypertext Transfer Secure Protocol (HTTPS) TCP Port 443: HTTPS GET Request
Observable 3 †	<i>Anomalous Network Traffic: From Local Host to External IP: Over File Transfer Protocol (FTP) TCP Port 22</i>
Observable 4 †	<i>Anomalous Network Traffic: From Local Host to External IP: Over Secure File Transfer Protocol (SFTP) TCP Port 23</i>
Observable 5	Anomalous Network Traffic: From External IP to Local Host: Over Hypertext Transfer Protocol (HTTP) TCP Port 80: HTTP POST Request
Observable 6	Anomalous Network Traffic: From External IP to Local Host: Over Hypertext Transfer Secure Protocol (HTTPS) TCP Port 443: HTTPS POST Request
Observable 7	Anomalous Network Traffic: From External IP to Local Host: Over File Transfer Protocol (FTP) TCP Port 22
Observable 8	Anomalous Network Traffic: From External IP to Local Host: Over Secure File Transfer Protocol (SFTP) TCP Port 23
Observable 9 †	<i>Anomalous Network Traffic: From External IP to Local Host: Over Server Message Block (SMB) TCP Port 445</i>
Observable 10 †	<i>Anomalous Network Traffic: From External IP to Local Host: Over Remote Desktop Protocol (RDP) TCP Port 3389</i>

Observables Associated with Commonly Used Port Technique (T0855)

Observable 1	Anomalous Network Traffic: From Local Host to External IP: Over Hypertext Transfer Protocol (HTTP) TCP Port 80: HTTP GET Request
Observable 2	Anomalous Network Traffic: From Local Host to External IP: Over Hypertext Transfer Secure Protocol (HTTPS) TCP Port 443: HTTPS GET Request
Observable 3 †	<i>Anomalous Network Traffic: From Local Host to External IP: Over File Transfer Protocol (FTP) TCP Port 22</i>

Observables Associated with Commonly Used Port Technique (T0855)

Observable 4 †	<i>Anomalous Network Traffic: From Local Host to External IP: Over Secure File Transfer Protocol (SFTP) TCP Port 23</i>
Observable 5	<i>Anomalous Network Traffic: From External IP to Local Host: Over Hypertext Transfer Protocol (HTTP) TCP Port 80: HTTP POST Request</i>
Observable 6	<i>Anomalous Network Traffic: From External IP to Local Host: Over Hypertext Transfer Secure Protocol (HTTPS) TCP Port 443: HTTPS POST Request</i>
Observable 7	<i>Anomalous Network Traffic: From External IP to Local Host: Over File Transfer Protocol (FTP) TCP Port 22</i>
Observable 8	<i>Anomalous Network Traffic: From External IP to Local Host: Over Secure File Transfer Protocol (SFTP) TCP Port 23</i>
Observable 9 †	<i>Anomalous Network Traffic: From External IP to Local Host: Over Server Message Block (SMB) TCP Port 445</i>
Observable 10 †	<i>Anomalous Network Traffic: From External IP to Local Host: Over Remote Desktop Protocol (RDP) TCP Port 3389</i>

Observables Associated with Remote Services Technique (T0886)

Observable 1 †	<i>Anomalous Network Traffic: From External IP to Local Host: Over Server Message Block (SMB) TCP Port 445</i>
Observable 2 †	<i>Anomalous Network Traffic: From External IP to Local Host: Over Remote Desktop Protocol (RDP) TCP Port 3389</i>
Observable 3	<i>Anomalous Host Activity: Successful Logon From External Host: Valid User Account Windows Event ID (4624): Anomalous timestamp</i>
Observable 4	<i>Anomalous Host Activity: Successful Logon From External Host: Valid User Account Windows Event ID (4624): Anomalous remote IP</i>

Observables Associated with Valid Accounts Technique (T0859)

Observable 1 †	<i>Anomalous Network Traffic: From External IP to Local Host: Over Server Message Block (SMB) TCP Port 445</i>
Observable 2 †	<i>Anomalous Network Traffic: From External IP to Local Host: Over Remote Desktop Protocol (RDP) TCP Port 3389</i>
Observable 3	<i>Anomalous Host Activity: Successful Logon From External Host: An Account Was Successfully Logged On (Windows Event ID 4624): Anomalous Timestamp</i>
Observable 4	<i>Anomalous Host Activity: Successful Logon From External Host: An Account Was Successfully Logged On (Windows Event ID 4624): Anomalous Remote IP</i>
Observable 5	<i>Anomalous Host Activity: Increase in Failed Login Attempts: An Account Failed To Log On (Windows Event ID 4625): Anomalous Timestamp</i>
Observable 6	<i>Anomalous Host Activity: Increase in Failed Login Attempts: An Account Failed To Log On (Windows Event ID 4625): Anomalous Remote IP</i>

Observables Associated with Lateral Tool Transfer Technique (T0867)	
Observable 1	Anomalous Network Traffic: From Local Host to External IP: Over Hypertext Transfer Protocol (HTTP) TCP Port 80: HTTP GET Request
Observable 2	Anomalous Network Traffic: From Local Host to External IP: Over Hypertext Transfer Secure Protocol (HTTPS) TCP Port 443: HTTPS GET Request
Observable 3 †	<i>Anomalous Network Traffic: From Local Host to External IP: Over File Transfer Protocol (FTP) TCP Port 22</i>
Observable 4 †	<i>Anomalous Network Traffic: From Local Host to External IP: Over Secure File Transfer Protocol (SFTP) TCP Port 23</i>
Observable 5	Anomalous Network Traffic: From Local Host to External IP: Via Rclone Application
Observable 6 †	<i>Anomalous Network Traffic: From Local Host to External IP: Over TCP Port 445</i>
Observable 7	Anomalous Network Traffic: From External IP to Local Host: Over Hypertext Transfer Protocol (HTTP) TCP Port 80: HTTP POST Request
Observable 8	Anomalous Network Traffic: From External IP to Local Host: Over Hypertext Transfer Secure Protocol (HTTPS) TCP Port 443: HTTPS POST Request
Observable 9 †	<i>Anomalous Network Traffic: From External IP to Local Host: Over File Transfer Protocol (FTP) TCP Port 22</i>
Observable 10 †	<i>Anomalous Network Traffic: From External IP to Local Host: Over Secure File Transfer Protocol (SFTP) TCP Port 23</i>
Observable 11 †	<i>Anomalous Network Traffic: From External IP to Local Host: Over Server Message Block (SMB) TCP Port 445</i>
Observable 12 †	<i>Anomalous Command Line: rclone copy <source:sourcepath> <dest:destpath></i>
Observable 13 †	<i>Anomalous Command Line: mega-export -a <Local Host File Path> <Remote Host File Path></i>
Observable 14	Anomalous Command Line: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
Observable 15	Anomalous Binary on Local Host: rclone.exe
Observable 16	Anomalous Binary on Local Host: MEGAcmdShell.exe
Observable 17	Anomalous Binary on Local Host: empire.exe
Observable 18	Anomalous Binary on Local Host: koadic.exe
Observable 19	Execution of Anomalous Binary on Host: rclone.exe
Observable 20	Execution of Anomalous Binary on Host: MEGAcmdShell.exe
Observable 21	Execution of Anomalous Binary on Host: empire.exe
Observable 22	Execution of Anomalous Binary on Host: koadic.exe

Observables Associated with Command-Line Interface Technique (T0807)	
Observable 1	Anomalous Network Traffic: From Local Host to External IP: Over Hypertext Transfer Protocol (HTTP) TCP Port 80: HTTP GET Request

Observables Associated with Command-Line Interface Technique (T0807)	
Observable 2	Anomalous Network Traffic: From Local Host to External IP: Over Hypertext Transfer Secure Protocol (HTTPS) TCP Port 443: HTTPS GET Request
Observable 3 †	<i>Anomalous Network Traffic: From Local Host to External IP: Over File Transfer Protocol (FTP) TCP Port 22</i>
Observable 4 †	<i>Anomalous Network Traffic: From Local Host to External IP: Over Secure File Transfer Protocol (SFTP) TCP Port 23</i>
Observable 5	Anomalous Network Traffic: From Local Host to External IP: Via Rclone application
Observable 6	Anomalous Network Traffic: From Local Host to External IP: Over TCP Port 445
Observable 7 †	<i>Anomalous Command Line: rclone.exe copy <source:sourcepath> <dest:destpath></i>
Observable 8 †	<i>Anomalous Command Line: MEGAcmdShell.exe mega-export -a <Local Host File Path> <Remote Host File Path></i>
Observable 9	Anomalous Command Line: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
Observable 10	Anomalous Binary on Local Host: MEGAcmdShell.exe
Observable 11	Anomalous Binary on Local Host: empire.exe
Observable 12	Anomalous Binary on Local Host: koadic.exe
Observable 13	Execution of Anomalous Binary on Host: rclone.exe
Observable 14	Execution of Anomalous Binary on Host: MEGAcmdShell.exe
Observable 15	Execution of Anomalous Binary on Host: empire.exe
Observable 16	Execution of Anomalous Binary on Host: koadic.exe

Observables Associated with Theft of Operational Information Technique (T0882)	
Observable 1	Anomalous Network Traffic: From Local Host to External IP: Over Hypertext Transfer Protocol (HTTP) TCP Port 80: HTTP GET Request
Observable 2	Anomalous Network Traffic: From Local Host to External IP: Over Hypertext Transfer Secure Protocol (HTTPS) TCP Port 443: HTTPS GET Request
Observable 3 †	<i>Anomalous Network Traffic: From Local Host to External IP: Over File Transfer Protocol (FTP) TCP Port 22</i>
Observable 4 †	<i>Anomalous Network Traffic: From Local Host to External IP: Over Secure File Transfer Protocol (SFTP) TCP Port 23</i>
Observable 5	Anomalous Network Traffic: From Local Host to External IP: Via Rclone application
Observable 6 †	<i>Anomalous Network Traffic: From Local Host to External IP: Over TCP Port 445</i>
Observable 7 †	<i>Anomalous Command Line: rclone.exe copy <source:sourcepath> <dest:destpath></i>

Observables Associated with Theft of Operational Information Technique (T0882)

Observable 8 †	<i>Anomalous Command Line: MEGAcmdShell.exe mega-export -a <Local Host File Path> <Remote Host File Path></i>
Observable 9	Anomalous Command Line: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
Observable 10	Anomalous Binary on Local Host: rclone.exe
Observable 11	Anomalous Binary on Local Host: MEGAcmdShell.exe
Observable 12	Anomalous Binary on Local Host: empire.exe
Observable 13	Anomalous Binary on Local Host: koadic.exe
Observable 14	Execution of Anomalous Binary on Host: rclone.exe
Observable 15	Execution of Anomalous Binary on Host: MEGAcmdShell.exe
Observable 16	Execution of Anomalous Binary on Host: empire.exe
Observable 17	Execution of Anomalous Binary on Host: koadic.exe

Observables Associated with Masquerading Technique (T0849)

Observable 1	Presence of Anomalous Binary on Host: Signed Executable: Spoofed signature: Logitech Plug-In: SpotLife WebAlbum Service Plugin
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Observables Associated with Service Stop Technique (T0881)

Observable 1	Anomalous Command Line: C:\Windows\system32\takeown.exe /F <service_name>
Observable 2	Anomalous Command Line: C:\Windows\system32\icacls.exe <service_name> /reset
Observable 3 †	<i>Anomalous Host Activity: Windows Service Disabled: Windows Security Services</i>
Observable 4	Anomalous Host Activity: Windows Service Disabled: Email Services
Observable 5	Anomalous Host Activity: Windows Service Disabled: Backup Services
Observable 6	Anomalous Host Activity: Windows Service Disabled: Database Services
Observable 7 †	<i>Anomalous Host Activity: Windows Service Disabled: AntiVirus Services</i>
Observable 8	Anomalous Host Activity: Windows Process Terminated (Event ID 4689)
Observable 9 †	<i>Anomalous Host Activity: Windows Process Terminated: Windows Security Service Processes</i>
Observable 10	Anomalous Host Activity: Windows Process Terminated: Email Service Processes
Observable 11	Anomalous Host Activity: Windows Process Terminated: Backup Service Processes
Observable 12	Anomalous Host Activity: Windows Process Terminated: Database Service Processes

Observables Associated with Service Stop Technique (T0881)	
Observable 13 †	<i>Anomalous Host Activity: Windows Process Terminated: AntiVirus Service Processes</i>
Observable 14	Presence of Anomalous Binary on Local Host: kprocesshacker.sys

Observables Associated with Native API Technique (T0834)	
Observable 1	Anomalous Execution of Native OS API: Windows API: ZwTerminateProcess
Observable 2	Anomalous Execution of Native OS API: Windows API: NtCreateFile
Observable 3	Anomalous Execution of Native OS API: Windows API: ZwDeleteFile
Observable 4	Presence of Anomalous Binary on Host: C:\Users\<username>\Desktop\p1q135no.exe
Observable 5	Presence of Anomalous Binary on Host: C:\Users\<username>\Desktop\DoppelPaymer.exe
Observable 6	Presence of Anomalous Binary on Host: kprocesshacker.sys
Observable 7	Execution of Anomalous Binary on Host: C:\Users\<username>\Desktop\p1q135no.exe
Observable 8	Execution of Anomalous Binary on Host: Locator.exe
Observable 9	Execution of Anomalous Binary on Host: kprocesshacker.sys
Observable 10	Anomalous Execution of Binary on Host: ntdll.dll
Observable 11	Anomalous Execution of Binary on Host: kernel32.dll
Observable 12	Anomalous Execution of Binary on Host: advapi32.dll
Observable 13	Anomalous Execution of Binary on Host: shlwapi.dll
Observable 14	Anomalous Execution of Binary on Host: crypt32.dll

Observables Associated with Device Restart/Shutdown Technique (T0816)	
Observable 1	Anomalous Host Activity: Host reboots (Event ID 4609)
Observable 2 †	<i>Anomalous Host Activity: Host reboots: Reboots into Safe Mode: Displays Anomalous Text on Reboot</i>

Observables Associated with Data Destruction Technique (T0809)	
Observable 1	Anomalous Increase in System Resource Utilization: Increase in CPU Utilization
Observable 2	Anomalous Increase in System Resource Utilization: Increase in Hard Drive Activity
Observable 3	Anomalous Increase in System Resource Utilization: Increase in Network Activity
Observable 4	Anomalous Command Line: C:\Windows\system32\vssadmin.exe Delete Shadows /All /Quiet

Observables Associated with Data Destruction Technique (T0809)	
Observable 5	Anomalous Command Line: C:\Windows\system32\diskshadow.exe /s C:\Users\User\AppData\Local\Temp\<random>.tmp
Observable 6	Anomalous Command Line: C:\Windows\system32\takeown.exe /F <file>
Observable 7	Anomalous Command Line: C:\Windows\system32\icacls.exe <file> /reset
Observable 8	Anomalous deletion of data: Deletion of Windows Shadow Volume
Observable 9 †	<i>Anomalous Modification of Files: .doppeled Appended to Filenames</i>
Observable 10 †	<i>Anomalous Modification of Files: Files Encrypted on Host</i>
Observable 11 †	<i>Presence of Anomalous File: Anomalous File on Desktop: Howtodecrypt.txt</i>
Observable 12 †	<i>Presence of Anomalous File: Anomalous File on Desktop: Howtodecrypt.txt: File Contains TOR Link</i>
Observable 13 †	<i>Presence of Anomalous File: Anomalous File on Desktop: Howtodecrypt.txt: File Contains @protonmail.com Email Address</i>
Observable 14	Presence of Anomalous File: <random>.tmp file in %TEMP% folder
Observable 15	Presence of Anomalous File: Unencrypted System Volume Information
Observable 16	Presence of Anomalous File: Unencrypted \$RECYCLE.BIN
Observable 17	Presence of Anomalous File: Unencrypted WebCache

Observables Associated with Manipulation of View Technique (T0832)	
Observable 1 †	<i>Anomalous Modification of Files: .doppeled Appended to Filenames</i>
Observable 2 †	<i>Anomalous Modification of Files: Files Encrypted on Host</i>
Observable 3 †	<i>Presence of Anomalous File: Anomalous File on Desktop: Howtodecrypt.txt</i>
Observable 4 †	<i>Presence of Anomalous File: Anomalous File on Desktop: Howtodecrypt.txt: File Contains TOR Link</i>
Observable 5 †	<i>Presence of Anomalous File: Anomalous File on Desktop: Howtodecrypt.txt: File Contains @protonmail.com Email Address</i>
Observable 6	Anomalous Host Activity: Host Reboots (Event ID 4609)
Observable 7	Anomalous Host Activity: Host Reboots: Reboots into Safe Mode: Displays Anomalous Text on Reboot

Observables Associated with Loss of Productivity and Revenue Technique (T0882)	
Observable 1 †	<i>Anomalous Loss of Productivity: Business Processes Inaccessible: Enterprise Functionality</i>
Observable 2 †	<i>Anomalous Loss of Productivity: Business Processes Inaccessible: Standard Business Operations</i>
Observable 3 †	<i>Anomalous Loss of Productivity: Business Processes Inaccessible: Billing</i>

APPENDIX B: ARTIFACTS LIBRARY

Artifacts Associated with Spearphishing Attachment Technique (T0865)	
Artifact 1	Email .ost File
Artifact 2	Mismatched MIME and Attachment File Extension
Artifact 3	Email Sender Address
Artifact 4	Email Message
Artifact 5	Email Receiver
Artifact 6	Email Receiver Name
Artifact 7	Email Receiver Domain
Artifact 8	Email Receiver Address
Artifact 9	Enable Macros Pop-Up
Artifact 10	Email Application Log File
Artifact 11	Email Unified Audit Log File
Artifact 12	Email Service Name
Artifact 13	Suspicious Email Message Content
Artifact 14	Email Sender Domain
Artifact 15	Email .pst File
Artifact 16	Email Sender IP Address
Artifact 17	Simple Mail Transfer Protocol SMTP Traffic
Artifact 18	Mail Transfer Agent Logs
Artifact 19	Email Parent Process
Artifact 20	Mail Transfer Agent Logs
Artifact 21	Email Domain Name System DNS Traffic
Artifact 22	Email Domain Name System DNS Event
Artifact 23	File Attachment Warning Prompt
Artifact 24	Email Timestamp
Artifact 25	Email Attachment
Artifact 26	Email Attachment File Type
Artifact 27	Email Header
Artifact 28	Email Sender Name
Artifact 29	Operating System Service Creation

Artifacts Associated with Drive-by Compromise Technique (T0817)	
Artifact 1	Destination IP Address
Artifact 2	Industrial Application Disk Write

Artifacts Associated with Drive-by Compromise Technique (T0817)	
Artifact 3	Industrial Application Process
Artifact 4	Website
Artifact 5	TLS Certificates
Artifact 6	Disk Write
Artifact 7	Disk Read
Artifact 8	Application Log
Artifact 9	File Creation
Artifact 10	Source IP Address
Artifact 11	POWERSHELL Log Creation
Artifact 12	POWERSHELL Cmdlet Open
Artifact 13	Dialog Boxes Open
Artifact 14	cmd.exe Application Start
Artifact 15	Memory Evidence
Artifact 16	HTTP Traffic
Artifact 17	Child Processes Created
Artifact 18	Process Ending
Artifact 19	Process Creation
Artifact 20	SMB Traffic
Artifact 21	HTTPS Traffic
Artifact 22	DNS Traffic
Artifact 23	.lnk Files
Artifact 24	Prefetch Files

Artifacts Associated with User Execution Technique (T0863)	
Artifact 1	Command Execution
Artifact 2	Service Termination
Artifact 3	File Changes
Artifact 4	Increased ICMP Traffic (Network Scanning)
Artifact 5	Network Traffic Changes
Artifact 6	Application Installation
Artifact 7	Network Connection Creation
Artifact 8	Application Log Content
Artifact 9	User Account Modification
Artifact 10	File Creation

Artifacts Associated with User Execution Technique (T0863)

Artifact 11	Process Creation
Artifact 12	System Log
Artifact 13	Process Termination
Artifact 14	File Execution
Artifact 15	Prefetch Files
Artifact 16	Registry Modification
Artifact 17	File Modifications
Artifact 18	File Renaming
Artifact 19	System Patches Installed
Artifact 20	Files Opening
Artifact 21	File Signature Validation
Artifact 22	Installers Created
Artifact 23	Application Log

Artifacts Associated with Scripting Technique (T0853)

Artifact 1	Startup Menu Modification
Artifact 2	OS Service Installation
Artifact 3	Registry Modifications
Artifact 4	Network Services Created
Artifact 5	External Network Connections
Artifact 6	Prefetch Files Created
Artifact 7	Executable Files
Artifact 8	System Processes Created
Artifact 9	OS Timeline Event
Artifact 10	System Event Log Creation
Artifact 11	Files Dropped into Directory
Artifact 12	Windows Api Event Log

Artifacts Associated with Standard Application Layer Protocol Technique (T0869)

Artifact 1	SMB Traffic Port
Artifact 2	Network Connection Times
Artifact 3	External IP Addresses
Artifact 4	External Network Connections
Artifact 5	DNS Autonomous System Number

Artifacts Associated with Standard Application Layer Protocol Technique (T0869)

Artifact 6	Increase in the Number of External Connections
Artifact 7	RDP Traffic Port
Artifact 8	HTTP Traffic Port
Artifact 9	DNS Traffic Port
Artifact 10	HTTP Post Request
Artifact 11	HTTPS Traffic Port
Artifact 12	Network Content Metadata

Artifacts Associated with Commonly Used Port Technique (T0885)

Artifact 1	Unexpected Process Usage of Common Port Observed via Firewall Logs
Artifact 2	Unexpected Process Usage of Common Port Observed via OS Commands (netstat)
Artifact 3	Unexpected Process Usage of Common Port Observed via Memory
Artifact 4	Unexpected Process Usage of Common Port Observed via OS Logs
Artifact 5	Unexpected Host Communicating with Common Port On Industrial Asset

Artifacts Associated with Remote Services Technique (T0886)

Artifact 1	Mouse Movement
Artifact 2	Authentication Logs
Artifact 3	Network Traffic Content Creation
Artifact 4	Remote Session Creation Timestamp
Artifact 5	Process Creation
Artifact 6	VNC Traffic
Artifact 7	SMB Traffic
Artifact 8	SSH Traffic
Artifact 9	MSSQL Traffic 1433 Port
Artifact 10	File Movement
Artifact 11	Desktop Prompt Windows Created
Artifact 12	GUI Modifications
Artifact 13	System Log Event
Artifact 14	RDP Traffic
Artifact 15	Application Log
Artifact 16	Session Cache
Artifact 17	Unexpected
Artifact 18	Registry Connection Change

Artifacts Associated with Remote Services Technique (T0886)	
Artifact 19	Registry Changes
Artifact 20	Logoff Event
Artifact 21	Logoff
Artifact 22	Logon Event
Artifact 23	Remote Client Connection
Artifact 24	Data File Size In Network Content

Artifacts Associated with Valid Accounts Technique (T0859)	
Artifact 1	Logon Session Creation
Artifact 2	User Account Creation
Artifact 3	Logon Type Entry
Artifact 4	Logon Timestamp
Artifact 5	Failed Logon Event
Artifact 6	Successful Logon Event
Artifact 7	System Logs
Artifact 8	Default Credential Use
Artifact 9	Authentication Creation
Artifact 10	Prefetch Files Created After Execution
Artifact 11	Logons
Artifact 12	Application Log
Artifact 13	Domain Permission Requests
Artifact 14	Permission Elevation Requests
Artifact 15	Application Use Times
Artifact 16	Configuration Changes

Artifacts Associated with Lateral Tool Transfer Technique (T0867)	
Artifact 1	Remote Network Traffic
Artifact 2	File Metadata Changes
Artifact 3	User Information Changes
Artifact 4	Process Creation
Artifact 5	System Resource Usage Management Events
Artifact 6	Data Sent from One Location to Another
Artifact 7	Data Received from One Location to Another
Artifact 8	SQL Commands

Artifacts Associated with Lateral Tool Transfer Technique (T0867)	
Artifact 9	SQL Create Commands
Artifact 10	SQL Insert Commands
Artifact 11	Command Prompt Dialog Box Open
Artifact 12	SMB Traffic
Artifact 13	.dll Injection into File Directory
Artifact 14	.dll Execution
Artifact 15	Common Network Traffic
Artifact 16	Command Execution
Artifact 17	Industrial Network Traffic
Artifact 18	File Creation
Artifact 19	File Modification
Artifact 20	File Deletion
Artifact 21	File Location Change
Artifact 22	POWERSHELL Dialog Box Open

Artifacts Associated with Command-Line Interface Technique (T0807)	
Artifact 1	Command Execution
Artifact 2	Application Log
Artifact 3	HTTP Traffic
Artifact 4	Telnet Traffic
Artifact 5	SSH Traffic
Artifact 6	VNC Traffic Port
Artifact 7	Process Creation
Artifact 8	Remote Connections
Artifact 9	Process Ending
Artifact 10	Script Execution
Artifact 11	User Account Logon
Artifact 12	User Account Privilege Change
Artifact 13	Logon Event
Artifact 14	Event Log Type
Artifact 15	Event Log Type
Artifact 16	Failed Logon Event
Artifact 17	Command-Line Memory Data
Artifact 18	cmd.exe Application Execution

Artifacts Associated with Command-Line Interface Technique (T0807)	
Artifact 19	RDP Traffic
Artifact 20	Industrial Application Execution
Artifact 21	POWERSHELL Cmdlet Application Execution
Artifact 22	Event ID 4103 POWERSHELL Command
Artifact 23	Event ID 4688 Command-Line Execution
Artifact 24	NTUSER Application Execution Entries
Artifact 25	External Network Connection

Artifacts Associated with Theft of Operational Information Technique (T0882)	
Artifact 1	Exfiltration of Endpoint Host Data (Spreadsheets, Diagrams, Documents, Configurations, etc.) via Standard Protocols
Artifact 2	Exfiltration from Database via Standard Queries
Artifact 3	Exfiltration of Endpoint Host Data (Spreadsheets, Diagrams, Documents, Configurations, etc.) via Industrial Protocols
Artifact 4	Exfiltration of Operational Info via Phishing

Artifacts Associated with Masquerading Technique (T0849)	
Artifact 1	Command-Line Execution
Artifact 2	Additional Functionality In Applications
Artifact 3	Applications Causing Unintended Actions
Artifact 4	Leetspeak File Creation
Artifact 5	File Modification
Artifact 6	Process Metadata Changes
Artifact 7	Common Application with Non-Native Child Processes
Artifact 8	Scheduled Job Metadata
Artifact 9	Services Metadata
Artifact 10	Service Creation
Artifact 11	Scheduled Job Modification
Artifact 12	Additional File Directories Created
Artifact 13	File Creation with Common Name
Artifact 14	Leetspeak User Metadata
Artifact 15	Warez Application Use

Artifacts Associated with Service Stop Technique (T0881)	
Artifact 1	Internal System Logs

Artifacts Associated with Service Stop Technique (T0881)	
Artifact 2	Alarm Event
Artifact 3	OS API Call
Artifact 4	Application Error Messages
Artifact 5	Process Error Messages
Artifact 6	Application Service Stop
Artifact 7	Registry Change HKLM\SYSTEM\CURRENTCONTROLSET\SERVICES
Artifact 8	OS Service Crash
Artifact 9	System Event Logs
Artifact 10	Application Event Logs
Artifact 11	System Resource Usage Manager Application Usage Change
Artifact 12	Command-Line System Argument
Artifact 13	Process Failure

Artifacts Associated with Native API Technique (T0834)	
Artifact 1	Alert Generated
Artifact 2	System Resource Usage Management Changes
Artifact 3	.dll Modifications
Artifact 4	Imports Hash Changed
Artifact 5	Files Created
Artifact 6	Processes Initiated
Artifact 7	Services Initiated
Artifact 8	SYSMON Events Created
Artifact 9	Performance Degradation
Artifact 10	Blue Screen
Artifact 11	Configuration Change
Artifact 12	Command Execution
Artifact 13	Industrial Protocol Command Packet
Artifact 14	Host Device Failure
Artifact 15	Industrial Network Traffic
Artifact 16	Device Reads
Artifact 17	Device I/O Image Table Manipulated
Artifact 18	Device Failure
Artifact 19	Systems Calls
Artifact 20	Device Performance Degradation

Artifacts Associated with Native API Technique (T0834)	
Artifact 21	Device Memory Modification
Artifact 22	Device Alarm
Artifact 23	Device Live Data Changes
Artifact 24	Alter Process Logic
Artifact 25	Memory Corruption

Artifacts Associated with Device Restart/Shutdown Technique (T0816)	
Artifact 1	Logon Events
Artifact 2	Process Alarm
Artifact 3	Memory Corruption
Artifact 4	Unauthorized Input
Artifact 5	Command Prompt Opened
Artifact 6	Hardware Failure
Artifact 7	Logoff Events
Artifact 8	Local Network Connections
Artifact 9	Significant Operational Data Changes
Artifact 10	Blue Screen
Artifact 11	Reboot Screen
Artifact 12	Network Command Packets
Artifact 13	Loss of Network Connection
Artifact 14	Process Environmental Changes
Artifact 15	Process Failure
Artifact 16	Process Application Event
Artifact 17	External Network Connections

Artifacts Associated with Data Destruction Technique (T0809)	
Artifact 1	Command-Line Arguments
Artifact 2	Files Moved to Recycle Bin
Artifact 3	Missing Files
Artifact 4	Host System Reboot Failure
Artifact 5	Process Logic Failure
Artifact 6	Event Log Creation
Artifact 7	System Call
Artifact 8	System Application Interruption

Artifacts Associated with Data Destruction Technique (T0809)	
Artifact 9	Device Failure
Artifact 10	Recovery Attempt Failure
Artifact 11	TFTP Port
Artifact 12	SFTP Port
Artifact 13	Memory Corruption
Artifact 14	Use of File Transfer Protocols
Artifact 15	SCP Port
Artifact 16	File Encryptions
Artifact 17	Non-Native Files
Artifact 18	External Network Connections
Artifact 19	Transient Device Connections
Artifact 20	Program Execution
Artifact 21	Telnet Port
Artifact 22	FTPS Port
Artifact 23	HTTP Port
Artifact 24	HTTPS Port
Artifact 25	Local Network Connections
Artifact 26	FTP Port
Artifact 27	SMB Port

Artifacts Associated with Manipulation of View Technique (T0832)	
Artifact 1	Modification of Operating System or the Installation of a Filter Driver Could Lead to Manipulations of Packet at the Kernel Level
Artifact 2	File System Modification Artifacts Might Be Associated with the Manipulation of View Attack Might Be Present on Disk
Artifact 3	A Rogue Proxy, Gateway, or Network Device in the Path of the Industrial Communications Could Manipulate Traffic
Artifact 4	Compromise and Manipulation of Data Storage Locations Used to Produce or Present Information to Operators
Artifact 5	Modification of Application Libraries or Dependencies as Seen with STUXNET DLL Hooking

Artifacts Associated with Loss of Productivity and Revenue Technique (T0828)	
Artifact 1	Loss of Confidence in a Safety System Due to Unreliability Might Result in a Risk Management Driven Shutdown of a Plant
Artifact 2	Wormable or Other Highly Propagating Malware Might Result in the Shutdown of a Plant to Prevent Ransomware or Other Destructive Attacks

Artifacts Associated with Loss of Productivity and Revenue Technique (T0828)	
Artifact 3	Extortion Attempts Might Lead to Reduced Operations Due to Potential Presence of Malicious Attackers
Artifact 4	Loss of Control of Critical Systems Due to Ransomware or Loss of Confidence Might Lead to a Degraded Productivity or Revenue Operating State
Artifact 5	File System Modification Artifacts Might Be Associated with the Loss of Productivity and Revenue Attack Might Be Present On Disk

APPENDIX C: OBSERVERS

This is a collection of standardized potential observers that work in operational technology organizations. It has been slightly modified by the CyOTE team from the [Job Role Groupings listed in the SANS ICS Job Role to Competency Level Poster](#) to communicate the categories of potential observers during cyber events.

Engineering  • Process Engineer • Electrical, Controls, and Mechanical Engineer • Project Engineer • Systems and Reliability Engineer • OT Developer • PLC Programmer • Emergency Operations Manager • Plant Networking • Control/Instrumentation Specialist • Protection and Controls • Field Engineer • System Integrator	Support Staff  • Remote Maintenance & Technical Support • Contractors (engineering) • IT and Physical Security Contractor • Procurement Specialist • Legal • Contracting Engineer • Insurance • Supply-chain Participant • Inventory Management/Lifecycle Management • Physical Security Specialist
Operations Technology (OT) Staff  • Operator • Site Security POC • Technical Specialists (electrical/mechanical/chemical) • ICS/SCADA Programmer	Information Technology (IT) Cybersecurity  • ICS Security Analyst • Security Engineering and Architect • Security Operations • Security Response and Forensics • Security Management (CSO) • Audit Specialist
Operational Technology (OT) Cybersecurity  • OT Security • ICS/SCADA Security	• Security Tester
Management  • Plant Manager • Risk/Safety Manager • Business Unit Management • C-level Management	Information Technology (IT) Staff  • Networking and Infrastructure • Host Administrator • Database Administrator • Application Development • ERP/MES Administrator • IT Management

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